Fall 1983

Problem 1 (Fa83). Evaluate $\displaystyle\int_0^\infty \operatorname{sech}^2 x \cos \lambda x \, \mathrm{d}x$ where λ is a real constant.

Problem 2 (Fa83, Su85). Let $M_{n\times n}(\mathbb{F})$ be the ring of $n \times n$ matrices over a field $\mathbb{F}$. For $n \geqslant 1$ does there exist a ring homomorphism from $M_{(n+1)\times(n+1)}(\mathbb{F})$ onto $M_{n\times n}(\mathbb{F})$?

Problem 3 (Fa83, Sp87). Let $f : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ be a function which is continuously differentiable and whose partial derivatives are uniformly bounded:

$$\left| \frac{\partial f}{\partial x_i}(x_1, \ldots, x_n) \right| \leqslant M$$

for all $(x_1, \ldots, x_n) \neq (0, \ldots, 0)$. Show that if $n \geqslant 2$, then f can be extended to a continuous function defined on all of $\mathbb{R}^n$. Show that this is false if $n = 1$ by giving a counterexample.

Problem 4 (Fa83). Prove or disprove (by giving a counterexample), the following assertion: Every infinite sequence $x_1, x_2, \ldots$ of real numbers has either a nondecreasing subsequence or a nonincreasing subsequence.

Problem 5 (Fa83, Sp96). Let A be the $n \times n$ matrix which has zeros on the main diagonal and ones everywhere else. Find the eigenvalues and eigenspaces of A and compute $\det A$.

Problem 6 (Su80, Fa83, Sp84, Fa87, Fa96, Fa07). How many zeros does the complex polynomial

$$3z^9 + 8z^6 + z^5 + 2z^3 + 1$$

have in the annulus $1 < |z| < 2$?

Problem 7 (Fa83). Let G be a finite group and suppose that $G \times G$ has exactly four normal subgroups. Show that G is simple and non-abelian.

Problem 8 (Fa83). Let A be a linear transformation on $\mathbb{R}^3$ whose matrix, relative to the usual basis for $\mathbb{R}^3$, is both symmetric and orthogonal. Prove that A is either plus or minus the identity, or a rotation by $180°$ about some axis in $\mathbb{R}^3$, or a reflection about some two-dimensional subspace of $\mathbb{R}^3$.

Problem 9 (Fa83). For which real values of p does the differential equation

$$y'' + 2py' + y = 3$$

admit solutions $y = f(x)$ with infinitely many critical points?

Problem 10 (Fa83). Let $f : [0, \infty) \to \mathbb{R}$ be a uniformly continuous function with the property that

$$\lim_{b\to\infty} \int_0^b f(x)\, dx$$

exists (as a finite limit). Show that

$$\lim_{x\to\infty} f(x) = 0$$

Problem 11 (Fa83, Fa84). Prove or supply a counterexample: If f and g are C^1 real valued functions on $(0, 1)$, if

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} g(x) = 0$$

if g and g' never vanish, and if

$$\lim_{x\to 0} \frac{f(x)}{g(x)} = c$$

then

$$\lim_{x\to 0} \frac{f'(x)}{g'(x)} = c$$

Problem 12 (Fa83). Let $r_1, r_2, \ldots, r_n$ be distinct complex numbers. Show that a rational function of the form

$$f(z) = \frac{b_0 + b_1 z + \cdots + b_{n-2} z^{n-2} + b_{n-1} z^{n-1}}{(z - r_1)(z - r_2)\cdots(z - r_n)}$$

can be written as a sum

$$f(z) = \frac{A_1}{z - r_1} + \frac{A_2}{z - r_2} + \cdots + \frac{A_n}{z - r_n}$$

for suitable constants $A_1, \ldots, A_n$.

Problem 13 (Fa83). 1. Let $u(t)$ be a real valued differentiable function of a real variable t which satisfies an inequality of the form

$$u'(t) \leqslant au(t) \qquad t \geqslant 0 \qquad u(0) \leqslant b$$

where a and b are positive constants. Starting from first principles, derive an upper bound for $u(t)$ for $t > 0$.

2. Let $x(t) = (x_1(t), x_2(t), \dots, x_n(t))$ be a differentiable function from $\mathbb{R}$ to $\mathbb{R}^n$ which satisfies a differential equation of the form

$$x'(t) = f(x(t))$$

where $f : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous function. Assuming that f satisfies the condition

$$\langle f(y), y\rangle \leqslant \|y\|^2 \qquad y \in \mathbb{R}^n$$

derive an inequality showing that the norm $\|x(t)\|$ grows, at most, exponentially.

Problem 14 (Fa83, Sp87, Fa99). Let V be a finite dimensional complex vector space and let A and B be linear operators on V such that $AB = BA$. Prove that if A and B can each be diagonalized, then there is a basis for V which simultaneously diagonalizes A and B.

Problem 15 (Fa83). 1. Let f be a complex function which is analytic on an open set containing the disc $|z| \leqslant 1$, and which is real valued on the unit circle. Prove that f is constant.

2. Find a nonconstant function which is analytic at every point of the complex plane except for a single point on the unit circle $|z| = 1$, and which is real valued at every other point of the unit circle.

Problem 16 (Fa83). Let $F(t) = \big(f_{ij}(t)\big)$ be an $n \times n$ matrix of continuously differentiable functions $f_{ij} : \mathbb{R} \to \mathbb{R}$, and let

$$u(t) = \operatorname{tr}\big(F(t)^3\big)$$

Show that u is differentiable and

$$u'(t) = 3 \operatorname{tr}\big(F(t)^2 F'(t)\big)$$

Problem 17 (Fa83, Sp24, Sp26).

1. Let R be a finite integral domain, i.e., a finite commutative ring with 1 and no zero divisors. Prove that R is a field.

2. Show that if p is the characteristic of R then the Frobenius map $F : R \to R$, $F(x) = x^p$ is an automorphism (and in particular is bijective).

Problem 18 (Fa83). Let $f, g : \mathbb{R} \to \mathbb{R}$ be smooth functions with $f(0) = 0$ and $f'(0) \neq 0$. Consider the equation $f(x) = tg(x)$, $t \in \mathbb{R}$.

1. Show that in a suitably small interval $|t| < \delta$, there is a unique continuous function $x(t)$ which solves the equation and satisfies $x(0) = 0$.

2. Derive the first order Taylor expansion of $x(t)$ about $t = 0$.

Problem 19 (Fa83, Fa86). Prove that if p is a prime number, then the polynomial

$$f(x) = x^{p-1} + x^{p-2} + \cdots + 1$$

is irreducible in $\mathbb{Q}[x]$.

Problem 20 (Fa83). Let m and n be positive integers, with $m < n$. Let $M_{m\times n}$ be the space of linear transformations of $\mathbb{R}^m$ into $\mathbb{R}^n$, considered as $n \times m$ matrices, and let L be the set of transformations in $M_{m\times n}$ which have rank m.

1. Show that L is an open subset of $M_{m\times n}$.

2. Show that there is a continuous function $T : L \to M_{n\times m}$ such that $T(A)A = I_m$ for all A, where I_m is the identity on $\mathbb{R}^m$.